

Assignment 1: Corpus Construction and Characterization

Group topic: AI-assisted legal case research

Subtopic: Intellectual property law (Federal Court of Australia, 2006–2009)

1. Human vs. SLM (Phi-4-mini-instruct)

I sampled 25 records at random from my 218-record corpus using seed 7, and labeled all of them by hand. The schema has four fields. *ip_area* and *outcome* are constrained enums. *cited_cases* and *legislation_cited* are open lists. I then ran the same 25 records through Phi-4-mini-instruct with the baseline prompt (v1_original) and greedy decoding, and validated every output against the generated JSON schema.

Field	Accuracy	Errors (of 25)	Precision	Recall	F1
ip_area	0.64	9	—	—	—
outcome	0.80	5	—	—	—
cited_cases	0.68	8	0.526	0.192	0.282
legislation_cited	0.48	13	0.559	0.559	0.559

All four fields correct on the same record: 0.24. Schema violations: 0.00. Mean latency: 20.9s.

Accuracy by modality: table 0.40, text 0.20.

I had to throw out an earlier run. My first attempt gave a 60% schema-violation rate, with 15 of 25 records returning nothing usable. When I looked at the raw output I expected to find JSON that had been cut off. Instead each failure was several hundred repetitions of the character "!". That is token id 0, which is what greedy decoding falls back to once the logits become NaN, so the model was not producing a bad answer, but it was not computing at all.

The failures turned out to depend on input length. Records that failed had a median prompt of 3,027 characters, while records that parsed had a median of 1,436. This is a known instability in the Apple MPS backend at longer sequence lengths. I tried float32, which did not fit in 16 GB of unified memory and made the machine swap, and float16, which failed the same way. What finally worked was reducing *MAX_TEXT_CHARS* from 3000 to 1400 so that every prompt stayed under the threshold. That is a workaround rather than a fix, and I discuss what it cost me in section 4.

Phi does reasonably well on the enum fields when the passage names the statute or the subject matter. Outcome reaches 0.80, mostly because it correctly returns *not_stated* on the many mid-judgment sections that do not decide anything. Table records score 0.40 against 0.20 for prose. That gap makes sense to me: a citation table puts the answer in a column, while a judgment section buries it in running text.

Two things go wrong repeatedly. The first is granularity on statute names. Where I wrote “*Federal Court of Australia Act 1976*”, Phi wrote “*s 56 of the Federal Court of Australia Act 1976*”. It read the statute correctly, but the strings just do not match under exact comparison, so it scores as an error.

The second is citation recall. Precision on *cited_cases* is 0.526 and recall is 0.192, so Phi rarely invents a case that is not in the document, but it only finds about a fifth of the ones that are. I should flag that the 1,400-character cap makes this worse than it looks, since I wrote my labels from the full 4,500-character record and Phi never saw the rest of it. Either way, the problem with this model on this corpus is that it misses citations, not that it makes them up.

Representative disagreements:

doc_id	Field	My label	Phi (v1)
ip_003_09_68_s2	legislation_cited	Federal Court of Australia Act 1976; Federal Court Rules	Federal Court of Australia Act 1976; s 56 of the Federal Court of Australia Act 1976
ip_004_06_440_s1	ip_area	passing_off	trade_mark
ip_004_06_440_s3	outcome	refused	not_stated
ip_005_08_1588_s2	ip_area	registered_design	copyright

These four are typical of the whole set. The first is the granularity problem: Phi found the right statute and then wrote it two different ways, one of which includes a section number my label does not. The second and fourth are genuine classification errors, and both are near misses in legal terms. Passing off and trade mark are pleaded together constantly, and the design case does discuss drawings, which is copyright territory. The third is the opposite failure to the one I expected: Phi is too cautious rather than too confident, returning *not_stated* where the passage does record a refusal.

2. Recovery Strategy

The recurring failure I picked was *legislation_cited*, which was my worst field at 13 errors out of 25. Reading through the disagreements, this is a convention mismatch rather than a comprehension problem. Phi kept attaching section numbers to Act names and my labels did not. Some of that is my fault: my schema said "copied exactly as written" without saying whether a section number or a jurisdiction suffix belonged in the string, so there was nothing for the model to match against.

I tested two revisions against the same v1 baseline, rerunning only the 19 records v1 got wrong.

Prompt v1_original (generated from taxonomy.json):

"You label documents from a corpus about intellectual property law (Federal Court of Australia, 2006-2009). Fill in every field:

- ip_area: the branch of intellectual property law this document concerns -- one of patent, trade_mark, copyright, registered_design, passing_off, not_stated.

- outcome: the result recorded in this document; use not_stated if the document does not say -- one of granted, refused, dismissed, allowed,

not_stated.

- *cited_cases*: names of earlier court cases cited in this document, copied exactly as written -- a list of strings.
- *legislation_cited*: names of Acts or statutory provisions named in this document, copied exactly as written -- a list of strings.”

Prompt v3_actonly (my revision; only the last bullet changes):

“You label documents from a corpus about intellectual property law (Federal Court of Australia). Fill in every field:

- *ip_area*: the branch of intellectual property law this document concerns -- one of *patent*, *trade_mark*, *copyright*, *registered_design*, *passing_off*, *not_stated*.
- *outcome*: the result recorded in this document -- one of *granted*, *refused*, *dismissed*, *allowed*, *not_stated*.
- *cited_cases*: names of earlier court cases cited in this document, copied exactly as written -- a list of strings.
- *legislation_cited*: names of Acts named in this document -- a list of strings. Give the full name of the Act as the document writes it, including the year and any jurisdiction suffix such as (Cth). Do not include section numbers: write “Trade Marks Act 1995 (Cth)”, not “s 41 of the Trade Marks Act 1995”. List each Act once.”

Field	v1 baseline	v2 generic strict	v3 targeted
ip_area	0.526	0.526	0.526
outcome	0.737	0.684	0.789
cited_cases	0.579	0.579	0.579
legislation_cited	0.316	0.368	0.421
records fully fixed	0	1	1

Accuracy on the 19 records that failed under v1.

v2 is the starter code's generated recovery prompt. It restates every field rule more forcefully, with phrases like **MUST** be exactly one of and never invent a value. It was written before I had looked at a single error, so it is not aimed at anything specific. *legislation_cited* went up a little to 0.368, *outcome* dropped to 0.684, and a new schema violation appeared that had not been there before. Nothing really improved.

v3 is the one I wrote myself. I added a single rule to *legislation_cited* telling the model to give the full name of the Act with the year and jurisdiction suffix, to leave out section numbers, and I included a worked example. On the field I was aiming at, it worked. Accuracy went from 0.316 to 0.421 and errors dropped from 13 to 11.

What I did not expect was the damage to a field I never mentioned. *ip_area* accuracy stayed at 0.526, but the answers collapsed. 14 of the 19 records came back as *trade_mark*, where v1 had

produced a spread of five different values. The only explanation I can see is my worked example, which used *Trade Marks Act 1995 (Cth)*. I put a concrete example in to fix formatting in one field and it ended up biasing an unrelated classification field.

So the targeted revision beat the generic one, which suggests it is worth reading your errors before rewriting a prompt. But I think the *ip_area* collapse is the more useful finding. Prompt changes in a multi-field schema are not local. For the kind of legal-research API our group is designing, that is a real problem, because adjusting a citation-formatting rule could quietly change how the system classifies the area of law and nothing would flag it.

3. Cluster Reading

The clustering runs k-means over TF-IDF vectors of *raw_text*. At k=6 I got a silhouette of 0.176 with 206 of 218 records in one cluster. At k=10 the silhouette only rose to 0.193 and 196 of 218 were still in one cluster. Raising k did not separate anything, so I stopped there and treated the result itself as the finding.

Cluster	n	Interpretation
0	196	Undifferentiated bulk - table records with no text, plus general FCA prose
1	2	Patent opposition and issue estoppel
2	3	Bottled-water get-up and trade dress
3	3	Rice-cake packaging and trade marks
4	2	Footwear shape marks
5	3	Road-barrier patent infringement
6	3	Luxury-apparel logo and copyright
7	3	House-plan copyright
8	1	Cross-claim pleadings
9	2	Garment design registration

What the clusters actually separate is individual disputes, not areas of law. Every term that distinguishes a cluster is case-specific: *termite*, *guardliner*, *polo*, *redberry*. None of them is doctrinal, and no cluster lines up with "patent cases" or "copyright cases".

I think there are two reasons for this. My corpus is lexically very uniform, because everything comes from the same court in the same jurisdiction over four years, so procedural words like applicant, respondent, Act and proceeding show up in every record and swamp the differences. On top of that, the clusterer only vectorizes *raw_text*, and my 68 table records have *raw_text* set to null. They all become the empty string, get identical vectors, and land in the same place. They make up 68 of the 196 records in cluster 0, so about a third of my corpus is effectively invisible here.

I could not make the comparison to Phi's performance work. Because I drew the evaluation sample at random from the whole corpus instead of stratifying by cluster, 21 of my 25 labeled records sit in cluster 0 and every other cluster has at most one. The per-cluster accuracies of 1.0 and 0.0 are single records, so they tell me nothing. The only contrast I can actually defend is by modality: 0.40 on table records against 0.20 on text.

This matters for what our group is building. The API is supposed to retrieve relevant precedent for a legal question, and what this suggests is that text similarity over judgment prose would retrieve cases about the same product rather than cases answering the same legal question. A bottled-water passing-off case and a footwear trade-mark case are close doctrinally and far apart lexically. Retrieval on raw judgment text would need some doctrinal signal that surface similarity does not give you.

4. Corpus Limitations

Everything I have comes from one source. All 218 records are Federal Court of Australia judgments from 2006 to 2009, so the corpus cannot answer anything about state courts, the High Court, other jurisdictions, or any law from the last fifteen years. A system built on it would be confidently wrong about current Australian IP law. My records are thirds of judgments capped at 4,500 characters and Phi only saw the first 1,400 of each, which means an argument that runs across a cut is split between records and no single record holds complete reasoning. There are only 50 cases in total, and registered design (4 cases) and passing off (5) are too thin for me to claim anything about those areas. The corpus also records what each judgment cited but not how later courts treated it, so it cannot tell you whether a precedent is still good law, which is probably the first thing a practitioner would want to know. The licensing is ambiguous as well: the dataset readme says research use only and no redistribution, while the repository page says CC BY 4.0. I followed the stricter terms.

5. Optional Bonus: SLM vs. LLM

I ran the same 25 records through *openai/gpt-oss-120b* on the Groq API, using the same schema and the same v1 prompt. That is roughly thirty times the parameters of Phi-4-mini-instruct, which is 3.8B. Both models saw the same 1,400-character inputs.

Metric	Phi-4-mini (3.8B)	gpt-oss-120b
all fields correct	0.24	0.36
schema violations	0.00	0.00
mean latency (s)	20.9	5.7
ip_area	0.64	0.84
outcome	0.80	0.88
cited_cases	0.68	0.68
legislation_cited	0.48	0.56

The bigger model is clearly better at classification. *ip_area* goes from 0.64 to 0.84 and *outcome* from 0.80 to 0.88, and it is four times faster despite being much larger, because it runs on dedicated inference hardware instead of my laptop. Both models produced valid JSON on every record.

What struck me is where scale did not help. *cited_cases* is identical at 0.68, and recall is still low at 0.269 against Phi's 0.192. *legislation_cited* only moves from 0.48 to 0.56, and the hosted model makes the same granularity error, returning “s 75(2) of the Designs Act” alongside “*Designs Act*”. Pulling every citation out of legal prose, and matching my formatting conventions, are apparently not problems that thirty times the parameters solves.

Reproducibility

Corpus built with `build_corpus.py` (50 cases, seed 0) from the Legal Case Reports dataset. Evaluation sample drawn with `make_human_labels.py --n 25 --seed 7`. Phi-4-mini-instruct, bfloat16 on MPS, greedy decoding, `max_new_tokens 512`, `MAX_TEXT_CHARS 1400`. Hosted comparison used `openai/gpt-oss-120b` via Groq. The `v1_original`, `v2_recovery` and `v3_actonly` prompts are all kept in `taxonomy.json` and in the recovery JSON files. I used generative AI (Claude) to write the collection and helper scripts and to draft this memo. The 25 human labels, the labeling conventions and the interpretation of the results are mine.

References

- *Dataset: UCI Machine Learning Repository*. (n.d.). Archive.Ics.Uci.Edu. Retrieved September 23, 2026, from <https://archive.ics.uci.edu/dataset/239/legal+case+reports>
- Help with the code: Anthropic. *Claude Opus 5*. Claude.Ai. <https://claude.ai/new>